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Cryptographic Chain-of-Custody for Last-Mile Parcel Delivery: A Blockchain–IoT System Architecture with Discrete-Event Simulation Evaluation

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Article

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• ORCID: YOUR-ORCID-HERE

Abstract: Last-mile delivery remains the most operationally complex, fraud-prone, and costly segment of modern supply chains. This paper proposes a three-tier blockchain–IoT system architecture that creates a cryptographic chain-of-custody from the physical edge device to an immutable ledger. An edge IoT cryptographic seal generates a "physical hash" by encrypting sensor telemetry, binding it to a timestamp, and applying SHA-256 to produce a tamper-evident composite digest signed with a device-held ECDSA private key. A smart contract layer verifies signatures and hashes, evaluates business rules, and records custody transitions. A discrete-event simulation of 50,000 custody handoffs demonstrates a tamper detection rate of 99.96%, with edge-optimized contract design reducing transaction costs by 25% while preserving sub-15-second confirmation latency for 94% of events on Ethereum mainnet analogs and sub-4-second median latency on Layer-2 configurations.

Keywords: blockchain; Internet of Things; cryptographic chain-of-custody; last-mile delivery; smart contracts; tamper detection; discrete-event simulation; EIP-1559; supply chain visibility

1. Introduction

The global supply chain is undergoing a structural transition from centralized tracking architectures to distributed, cryptographically secured frameworks. Last-mile delivery—the final movement from distribution hub to end consumer—concentrates disproportionate operational risk, with dynamic routes, frequent custody changes, and unsupervised physical interactions. Package theft cost U.S. consumers approximately $19.5 billion annually [3], while pharmaceutical supply chain fraud accounts for $200 billion in global losses [4].

Blockchain technology provides an immutable, append-only ledger replicated across a peer-to-peer network, eliminating single-administrator vulnerabilities. Internet of Things (IoT) devices provide continuous sensing at the physical edge. Their convergence enables a "hyperconnected logistics" paradigm where each custody handoff becomes a cryptographically verifiable state transition [5,6]. Despite substantial conceptual work, three gaps persist: (1) most frameworks rely on manual attestations [7]; (2) IoT tamper events are logged to centralized databases [8]; (3) few architectures quantify latency–cost tradeoffs [9]. [9] is about Ethereum EIP-1559; i.e., not about latency-cost in blockchain based supply chains

This paper presents a three-tier architecture evaluated through discrete-event simulation of 50,000 custody handoffs, organized around three research questions: (RQ1) How can IoT seals cryptographically prove custody handoffs? (RQ2) To what extent does on-chain verification reduce fraud versus centralized systems? (RQ3) What are the latency–cost tradeoffs under varying conditions?

2. Related Work

most references are kind of old; this is acceptable when discussing the problem potential and main frame work, but not when discussing its current stage (at 2020 is different than at 2026) or the order as a cited use case (at least prospect on current state)

2.1. Blockchain-Enabled Supply Chain Management

Kshetri (2018) and Saberi et al. (2019) articulated blockchain's potential to address information asymmetry and trust deficits in logistics [10,11]. Abeyratne and Monfared (2016) proposed manufacturing provenance on immutable ledgers [12]. Tian (2016) demonstrated agri-food traceability [13], Wang et al. (2019) extended to pharmaceutical cold chains [14], and Casino et al. (2019) identified supply chain as the second most-cited blockchain use case [15]. Pournader et al. (2020) concluded most architectures remain at proof-of-concept stage [16]. Critically, Cole et al. (2019) noted that manual attestation models reintroduce human trust dependencies [7].

From the sophisticated blockchain community:
This is a nice old reference, 2018, https://scet.berkeley.edu/blockchain-technology-beyond-cryptocurrencies-case-1-future-last-mile-delivery/
a newer 2026 one, https://www.leangroup.com/resources/how-blockchain-and-iot-are-revolutionizing-supply-chain-transparency

2.2. IoT Tamper Detection and Physical–Digital Anchoring

Xu et al. (2014) established IoT sensing taxonomies for freight monitoring [17]. Physical unclonable functions provide device-level authentication [19]. Hayward et al. (2022) reported >98% tamper detection using multi-sensor fusion [20]. Chronicled's CryptoSeal demonstrated physical seal–blockchain binding [21]. Fernández-Caramés and Fraga-Lamas (2018) and Dai et al. (2019) proposed conceptual IoT–blockchain convergence [5,6], yet few studies specify complete protocols where the IoT device initiates on-chain transactions. Tamper events remain logged to centralized cloud databases [8].

Again, the statement needs examples which should better be from more recent references Your [36] (not cited) discusses this, but 2017 is too old in the blockchain field.

2.3. Smart Contract Protocols for Custody Verification

Christidis and Devetsikiotis (2016) analyzed smart contract applicability to IoT [22]. Korpela et al. (2017) proposed replacing paper bills of lading [23]. Hasan et al. (2020) described pharmaceutical verification reliant on manual QR scans [24]. Omar et al. (2021)

This is a 2/2026 Sustainable development survey, https://onlinelibrary.wiley.com/doi/10.1002/sd.70745
1-As a basis of comparison, this paper must contain more detailed insight about blockchain usage, challenges,..etc.
2-Check tables 11 & 12 for recent references and current status:
Hamdan et al. (2023), Zhu et al. (2022, 2024), Monoarfa et al. (2024), and Vasić et al. (2024)

reported 40% dispute-time reduction with Hyperledger Fabric [26]. PackChain uses Ethereum for crowd-sourced delivery management [27,28] but emphasizes transaction settlement over continuous physical monitoring. No published work atomically binds device identity, sensor attestation, and party identities in a single transaction.

are you sure?

2.4. Last-Mile Verification and Cost–Latency Tradeoffs

Boyer et al. (2009) established that last-mile operations constitute up to 53% of shipping cost [29]. Blockchain-based alternatives include drone delivery [31], reputation-based crowdsourcing [32], and sustainable ecosystem models [33]. Zheng et al. (2018) benchmarked Ethereum at ~15 TPS [34]. Malik et al. (2020) demonstrated 90% gas reduction via hash-only on-chain storage [37]. Kim et al. (2021) estimated $0.50–$4.20 per-shipment costs [38]. Roughgarden's EIP-1559 analysis showed reduced fee volatility [9]. Table 1 positions this work relative to the literature.

this was in 2018, find newer TPS

if you are going to discuss these issues, then you must discuss the impact of POS since 2022, and L2 solutions. Also, should discuss that Hyperledger Fabric is generally more adjusted to supply chains needs; i.e., discuss and compare different blockchains used in the solution.

1-In general, the paper is relatively shorter than average journal papers; section 2 is a proper place where expansion is due.
2-Refs [18], [25],[30],[35],[36] are missing from the text

Table 1. Positioning of the present contribution relative to prior work.

Dimension	Prior Work	Present Contribution
Custody attestation	Manual scan / QR code	Autonomous IoT seal with ECDSA identity
Tamper evidence	Centralized cloud logging	On-chain immutable record via smart contract
Handoff protocol	Informal specification	Atomic three-party transaction
Last-mile scope	Limited or excluded	End-to-end: warehouse → carrier → consumer
Evaluation	Qualitative or absent	DES: 50,000 handoffs with TDR, latency, gas benchmarks

ref [21] (CryptoSeal) comparison details? needs more expansion.
The CCN article talks about pharmaceuticals, are there more details? github is about documents (https://github.com/kirtanpatel2201/CryptoSeal/tree/main)

The paper lacks more detailed definition, especially that it is considered a basis of comparison here ([33],[33/1-4])

more references:
Course or thesis in 2022/2023: https://tesi.luiss.it/36467/1/747921_FRANCESCANGELI_VITTORIA.pdf
IFAC 2022 paper: https://www.sciencedirect.com/science/article/pii/S2405896322021334

3. System Architecture

3.1. Three-Tier Overview

The system consists of three tiers: (1) IoT Cryptographic Seal Device—a low-power edge device with environmental sensors, geolocation, and secure cryptographic element; (2) Smart Contract Execution Layer—Turing-complete contracts on an EVM-compatible blockchain for verification and state recording; (3) Visibility API—a stateless translation layer exposing REST/GraphQL endpoints.

3.2. Tier 1: IoT Cryptographic Seal Device

Each parcel is instrumented with sensors measuring temperature (±0.1 °C), humidity (±2% RH), tri-axis shock (±16g, 100 Hz), GNSS/GPS (sub-5 m accuracy), ARM Cortex-M4 microcontroller, secure element (ATECC608B), and BLE 5.0/NFC/cellular connectivity. During handoff, the device: (1) collects telemetry; (2) authenticates the receiving party via QR/BLE/NFC; (3) encrypts via AES-256-GCM; (4) constructs a canonical message; (5) applies SHA-256 for composite digest; (6) signs with ECDSA (secp256k1) via the secure element.

3.3. Tier 2: Smart Contract Execution Layer

Three contracts comprise this tier. DeviceRegistry maps authorized device IDs to ECDSA public keys. CustodyVerifier executes: (a) device authentication; (b) ECDSA signature verification via ecrecover; (c) SHA-256 hash recomputation; (d) timestamp monotonicity validation; (e) sequence number validation; (f) business rule evaluation; (g) state recording with event emission. AssetLedger maintains complete custody history with Merkle proof generation.

3.4. Tier 3: Visibility API

Implemented with Web3.py, the API connects to blockchain nodes via JSON-RPC, subscribes to contract events, and exposes REST/GraphQL endpoints. Clients receive custody status, anomaly alerts, audit trails, and Merkle inclusion proofs. The API is stateless and never mutates on-chain state, preserving the trust model.

4. Cryptographic Protocol Design

4.1. Composite Hashing

For each event, the device constructs M = (E, T, C) where E is the encrypted payload, T the monotonic timestamp, and C contextual metadata. The composite digest H(M) = SHA-256(E || T || C) uses canonical length-prefixed serialization. SHA-256's avalanche property ensures any bit-level change produces a drastically different hash (collision resistance ~2^128).

4.2. Digital Signatures

The device signs: σ = Sign(sk_device, H(M)) using ECDSA over secp256k1. The CustodyVerifier executes: (a) Verify(σ, H(M), pk_device) via ecrecover; (b) recompute H′ = SHA-256(E || T || C); (c) assert H′ = H(M). Successful verification cryptographically attributes the handoff to a specific device at a specific time.

It appears like the author is designing a permissioned private blockchain for the application (these are your TX validation rules) this could be stated more clearly. The skew window here resembles the confirmation escrow time (6 blocks in Bitcoin for example) where the replay maps to double spending.

4.3. Replay-Attack Resistance

Three mechanisms prevent replay: (1) timestamp integration with ±300 s skew window; (2) monotonically increasing sequence numbers; (3) optional challenge–response nonces for high-value corridors.

5. Evaluation via Discrete-Event Simulation

5.1. Simulation Environment

The DES uses Python 3.9+, SimPy 4.0, Web3.py 6.0 on Ganache 7.0, and EIP-1559-calibrated gas models [9]. Topology: 5 distribution centers, 12 transit hubs, 48 couriers, configurable validators (4–32). Parcel arrivals follow Poisson processes ($\lambda \in \{10, 50, 100, 500, 1000\}$ parcels/hour). Each parcel undergoes 2–5 handoffs. Block times: 12 s (Ethereum analog), 2 s (L2), 0.4 s (high-throughput).

5.2. Security Results

Table 2 presents tamper detection rates across 50,000 handoffs with 10% adversarial injection.

Table 2. Tamper detection rate by attack category (n = 50,000; 10% adversarial rate).

Attack Category	Injected	Detected	TDR (%)
Payload modification	1250	1250	100.00
Timestamp manipulation	1000	1000	100.00
Signature forgery	1000	1000	100.00
Replay attack	1000	998	99.80
Sequence violation	750	750	100.00
Aggregate total	5000	4998	99.96

The two undetected replays occurred when network latency placed the replayed transaction within the skew window before the original was confirmed. Reducing the window to ±120 s achieved 100.00% TDR with 0.3% false-rejection increase. Centralized baseline under insider threat: 12.4% TDR. what is insider threat here? elaborate more on that.

5.3. Latency Results

Table 3. End-to-end confirmation latency (λ = 100 parcels/hour).

Configuration	Median (s)	P95 (s)	P99 (s)
Ethereum analog (12 s blocks)	14.2	28.6	42.1
Layer-2 rollup (2 s blocks)	3.8	7.2	11.4
High-throughput (0.4 s blocks)	1.1	2.3	3.9

On Ethereum analog, 94% of events confirmed within 15 s. Under high load (λ = 1000), Ethereum median degraded to 38.7 s; L2 and high-throughput maintained <10 s.

5.4. Gas Cost Results

Table 4. Gas consumption: naïve vs. edge-optimized design.

Function	Naïve (gas)	Optimized (gas)	Reduction (%)
anchorEvent	148,200	92,400	37.7
verifyEvent	67,800	58,300	14.0
updateAssetLedger	44,600	44,600	0.0
Total per handoff	260,600	195,300	25.1
USD @ 30 gwei	$2.34	$1.76	24.8
USD @ 5 gwei (L2)	$0.39	$0.29	25.6

At L2 fees, $0.29 per handoff is viable for high-value shipments. Batch Merkle root strategies reduce costs to $0.04–$0.08.

6. Discussion

The 99.96% TDR assumes uncompromised device keys and firmware. Modern secure elements (ATECC608B, Infineon OPTIGA, NXP SE050) provide robust resistance, but advanced adversaries may theoretically extract keys. Deployments require tamper-resistant enclosures, secure boot chains, and remote attestation.

Public blockchains offer strongest censorship resistance but expose metadata and incur volatile fees. Consortium ledgers provide higher throughput and data privacy but reintroduce governance centralization. Hybrid designs anchoring consortium state to a public chain offer compromise. As stated earlier, this deserves more than this 4 lines paragraph.

Operational integration requires minimal UX friction—ideally a single NFC tap. Incentive-compatible designs (collateral deposits, reputation staking, automated penalties) align behavior with cryptographic enforcement [27,33]. The same as above (deserves more details).

The architecture complements existing freight brokerage visibility platforms, addressing the severe visibility gap between enterprise platforms and small brokerages [40]. The TOE framework [41] and Bricolage theory [42] identify this as addressing acute resource poverty in technology adoption.

7. Conclusions

This paper presented a three-tier blockchain–IoT architecture for cryptographic chain-of-custody in last-mile delivery, achieving 99.96% tamper detection across 50,000 simulated handoffs. Edge-optimized design reduces per-handoff cost to $0.29 on L2 networks with 3.8 s median latency. Compared to centralized baselines, TDR improves from 12.4% to 99.96% under insider-threat conditions.

Future work includes: (1) prototype deployment on Ethereum Sepolia and Polygon Amoy testnets; (2) zero-knowledge proofs for privacy-preserving disclosure; (3) cross-chain interoperability modeling; (4) a 90-day field trial with a logistics partner covering ≥10,000 live handoffs.

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